Chapter 12. Support Vector Machines

Lecture 9 (hyperplanes, maximal margin classifier, support vector classifier, kernels, multiclass SVM, SVM vs logistic regression)

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1 Separating Hyperplanes

1.1 Motivation

In a two-class problem with predictors X_1, X_2, suppose the classes are *linearly separable*: a straight line puts all points of one class on one side and all of the other class on the other. A natural approach is to find the line that gives the *biggest separation*, the points being as far from it as possible. This is the basic idea of support vector classifiers. For the general two-class problem we try to find a plane that separates the classes in feature space; if we cannot, we get creative in two ways: we soften what "separates" means, and we enrich and enlarge the feature space so that separation becomes possible.

1.2 Hyperplanes

A hyperplane in p dimensions is a flat affine subspace of dimension $p-1$:

$$\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p = 0;$$

in $p = 2$ dimensions it is a line. With $f(X) = \beta_0 + \beta^\top X$, $f(X) > 0$ on one side and $f(X) < 0$ on the other. Coding $y_i = +1$ for one class and $y_i = -1$ for the other, $f(X) = 0$ is a **separating hyperplane** if $y_i f(x_i) > 0$ for all i; then classify x by $\operatorname{sign} f(x)$, and $|f(x)|$ measures confidence.

Theorem: distance to a hyperplane

The signed distance of x to $\{f = 0\}$ is $f(x)/\|\beta\|$; if $\|\beta\| = 1$, $f(x)$ itself is the signed distance.

Proof [Supplement]. β is normal to the hyperplane. For a point x_0 on it, the distance is the length of the projection of $x - x_0$ on $\beta/\|\beta\|$: $\beta^\top(x - x_0)/\|\beta\| = (f(x) - f(x_0))/\|\beta\| = f(x)/\|\beta\|$. ■

2 The Maximal Margin Classifier

If the data are separable there are infinitely many separating hyperplanes. The **margin** M is the minimum perpendicular distance from the observations to the hyperplane; the **optimal separating hyperplane** maximizes it:

$$\max_{\beta_0,\beta} M \quad \text{subject to} \quad \sum_{j=1}^{p} \beta_j^2 = 1, \qquad y_i(\beta_0 + \beta^\top x_i) \geq M \quad \forall i.$$

Theorem: equivalent convex problem

The problem is equivalent to

$$\min_{\beta_0,\beta} \frac{1}{2}\|\beta\|^2 \quad \text{subject to} \quad y_i(\beta_0 + \beta^\top x_i) \geq 1 \quad \forall i,$$

with margin $M = 1/\|\beta\|$.

Proof [Supplement]. Drop the norm constraint and write the condition as $y_i(\beta_0 + \beta^\top x_i)/\|\beta\| \geq M$ (distances, by the previous theorem). The scale of (β_0, β) is arbitrary, so fix it by $\|\beta\| = 1/M$; then the constraints become $y_i(\beta_0 + \beta^\top x_i) \geq 1$ and maximizing $M = 1/\|\beta\|$ is minimizing $\|\beta\|^2/2$, a convex quadratic program. ■

The solution depends only on the points lying on the margin, the **support vectors**: moving any other point (without crossing the margin) does not change the hyperplane.

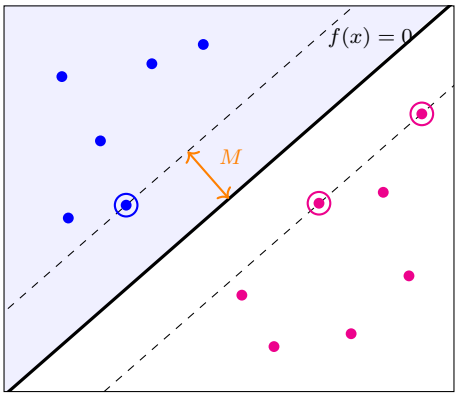


Figure 1: Maximal margin hyperplane (solid) and margins (dashed) for separable data. The circled points lie on the margins: they are the support vectors, and only they determine the hyperplane. M is the distance from the hyperplane to the margin (schematic).

2.1 Non-separable and noisy data

Often the data are not separable by a linear boundary (usually the case unless $n < p$), and then the problem has no solution. Even when separable, data may be noisy: a single observation can move the maximal margin hyperplane dramatically and produce a tiny margin. We want greater robustness to individual observations and better classification of *most* training observations: the support vector classifier maximizes a **soft margin**.

3 The Support Vector Classifier

Definition: soft-margin optimization

$$\max_{\beta_0,\beta,\epsilon} M \quad \text{subject to} \quad \sum_j \beta_j^2 = 1, \quad y_i(\beta_0 + \beta^\top x_i) \geq M(1-\epsilon_i), \quad \epsilon_i \geq 0, \quad \sum_{i=1}^n \epsilon_i \leq C.$$

ϵ_i are **slack variables**; $C \geq 0$ is a tuning parameter.

The slacks allow individual observations to violate the margin: $\epsilon_i = 0$ means observation i is on the correct side of the margin; $0 < \epsilon_i \leq 1$, on the wrong side of the margin but the correct side of the hyperplane; $\epsilon_i > 1$, on the wrong side of the hyperplane (misclassified). Since $\sum \epsilon_i \leq C$, at most C training observations can be misclassified.

C is a **budget** for violations. As C increases we tolerate more violations and the margin widens; as C decreases the margin narrows; $C = 0$ gives the maximal margin classifier. C is chosen by CV and controls the bias–variance trade-off: large C means many support vectors, low variance and possibly high bias; small C the opposite. Only observations on or inside the margin (or misclassified) are support vectors and affect the solution, so the classifier is robust to points far from the boundary — unlike LDA, which uses all points.

Derivation: the equivalent penalized form and the dual

In the parametrization of the previous section, the soft-margin problem is

$$\min_{\beta_0,\beta,\xi} \frac{1}{2}\|\beta\|^2 + \gamma \sum_i \xi_i \quad \text{s.t.} \quad y_i(\beta_0 + \beta^\top x_i) \geq 1 - \xi_i, \ \xi_i \geq 0,$$

where a *large* cost γ corresponds to a *small* budget C (software usually calls γ "C" — the

opposite role). The Lagrangian dual is

$$\max_{\alpha} \sum_i \alpha_i - \frac{1}{2}\sum_{i,i'} \alpha_i \alpha_{i'} y_i y_{i'} \langle x_i, x_{i'} \rangle \quad \text{s.t.} \quad 0 \le \alpha_i \le \gamma, \quad \sum_i \alpha_i y_i = 0,$$

and $\hat{\beta} = \sum_i \hat{\alpha}_i y_i x_i$. By complementary slackness $\hat{\alpha}_i > 0$ only for points with $y_i f(x_i) \le 1$, the support vectors. Hence $f(x) = \hat{\beta}_0 + \sum_{i \in \mathcal{S}} \hat{\alpha}_i y_i \langle x, x_i \rangle$, and the data enter only through inner products.

4 Support Vector Machines

4.1 Feature expansion

Sometimes a linear boundary won't work, whatever C. As in regression (interactions, polynomials, splines), enlarge the feature space with transformations, e.g. use $(X_1, X_2, X_1^2, X_2^2, X_1X_2)$ instead of (X_1, X_2). A linear boundary in the enlarged space,

$$\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_1^2 + \beta_4 X_2^2 + \beta_5 X_1 X_2 = 0,$$

is nonlinear (a conic) in the original space. With cubic polynomials the boundary is $\beta_0 + \cdots + \beta_6 X_1^3 + \beta_7 X_2^3 + \beta_8 X_1 X_2^2 + \beta_9 X_1^2 X_2 = 0$. Polynomials, especially high-dimensional ones, get wild rather fast; kernels introduce nonlinearities in a more elegant and controlled way.

4.2 Inner products and kernels

The inner product is $\langle x_i, x_{i'} \rangle = \sum_{j=1}^p x_{ij} x_{i'j}$. The linear support vector classifier can be written

$$f(x) = \beta_0 + \sum_{i=1}^n \alpha_i \langle x, x_i \rangle \qquad (n+1 \text{ parameters}),$$

and estimating (β_0, α) requires only the $\binom{n}{2}$ inner products between pairs of observations. Most $\hat{\alpha}_i$ are zero:

$$f(x) = \hat{\beta}_0 + \sum_{i \in \mathcal{S}} \hat{\alpha}_i \langle x, x_i \rangle,$$

where $\mathcal{S}$ is the support set, $\{i : \hat{\alpha}_i \neq 0\}$. If we can compute inner products between observations, we can fit an SV classifier; replacing $\langle x, x_i \rangle$ by a **kernel** $K(x, x_i)$ — an inner product $\langle \Phi(x), \Phi(x_i) \rangle$ in some feature space — gives the **support vector machine**

$$f(x) = \beta_0 + \sum_{i \in \mathcal{S}} \hat{\alpha}_i K(x, x_i).$$

- **Polynomial kernel** of degree d: $K(x_i, x_{i'}) = \left(1 + \sum_{j=1}^p x_{ij} x_{i'j}\right)^d$, which computes the inner products needed for d-dimensional polynomials.
- **Radial kernel**: $K(x_i, x_{i'}) = \exp\big(-\gamma \sum_{j=1}^p (x_{ij} - x_{i'j})^2\big)$, $\gamma > 0$: an implicit, infinite-dimensional feature space. Training points far from x contribute almost nothing, so the classifier is very local; γ controls the variance by squashing most dimensions severely (large γ, more flexible).

Example: the polynomial kernel is an inner product [Supplement]

For $p = 2$, $d = 2$: $(1+x^\top z)^2 = 1+2x_1z_1+2x_2z_2+x_1^2z_1^2+x_2^2z_2^2+2x_1x_2z_1z_2 = \langle\Phi(x), \Phi(z)\rangle$ with $\Phi(x) = (1, \sqrt{2}x_1, \sqrt{2}x_2, x_1^2, x_2^2, \sqrt{2}x_1x_2)$: the six features of a quadratic expansion, computed without forming them.

Kernel versus enlarged feature space. Computationally we only need $K(x_i, x_{i'})$ for the $\binom{n}{2}$ distinct pairs, whereas an explicit enlarged space can be intractable (the number of degree-d monomials grows like p^d), and some kernels (the radial kernel) correspond to no finite set of features. A function K is a valid kernel iff every Gram matrix $(K(x_i, x_{i'}))$ is positive semidefinite (Mercer).

Heart data. On the training set, ROC curves show the SV classifier slightly better than LDA, and radial SVMs with larger γ nearly perfect — overfitting; on the 90 test observations the radial SVM with large γ is worst, while the SV classifier, LDA and the radial SVM with small γ perform similarly.

5 SVMs With More Than Two Classes

The SVM as defined works for $K = 2$. Two popular extensions:

- **One versus all (OVA)**: fit K two-class SVMs $\hat{f}_k(x)$, each class versus the rest; classify x to the class with the largest $\hat{f}_k(x)$.
- **One versus one (OVO)**: fit all $\binom{K}{2}$ pairwise classifiers $\hat{f}_{kl}(x)$; classify x to the class that wins the most pairwise competitions.

If K is not too large, use OVO (each fit uses only two classes and is balanced).

6 SVM versus Logistic Regression

Motivation. The SVM criterion can be rewritten as loss plus penalty with the hinge loss, which is close to the logistic loss. This explains why the two methods often give similar results.

Theorem: SVM as loss plus penalty

The support vector classifier with $f(X) = \beta_0 + \beta_1X_1 + \cdots + \beta_pX_p$ solves

$$\min_{\beta_0,\beta} \sum_{i=1}^{n} \underbrace{\max[0,\, 1 - y_i f(x_i)]}_{\text{hinge loss}} + \lambda \sum_{j=1}^{p} \beta_j^2,$$

with large λ corresponding to a large budget C.

Proof [Supplement]. In the penalized form, at the optimum each slack equals $\xi_i = \max[0, 1 - y_i f(x_i)]$ (the smallest feasible value). Substituting gives $\frac{1}{2}\|\beta\|^2 + \gamma \sum_i \max[0, 1 - y_i f(x_i)]$; divide by γ and set $\lambda = 1/(2\gamma)$. ■

This is the "loss + penalty" form, like ridge regression. The hinge loss is exactly zero for observations on the correct side of the margin ($y_i f \geq 1$), which is why only support vectors matter. It is very similar to the logistic-regression loss (negative log-likelihood) $\log(1+e^{-y_i f(x_i)})$, which is never exactly zero but small for $y_i f \gg 0$; hence the two often give similar results.

Which to use?

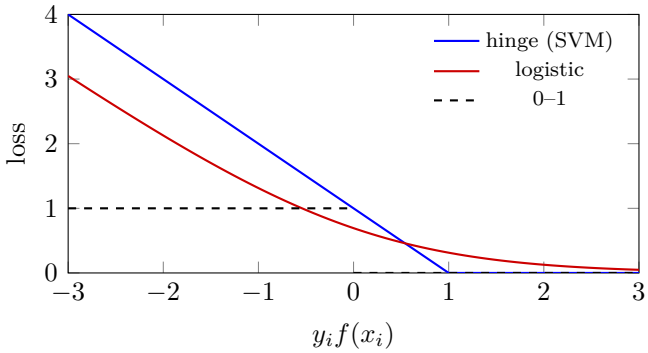


Figure 2: Hinge loss $\max(0, 1 - yf)$ is exactly zero once a point is beyond the margin ($yf \geq 1$); the logistic loss $\log(1 + e^{-yf})$ is a smooth version that never reaches zero. Both are convex surrogates of the 0–1 loss.

- When the classes are (nearly) separable, SVM does better than logistic regression; so does LDA.
- When not, logistic regression (with a ridge penalty) and SVM are very similar.
- To estimate probabilities, logistic regression is the choice.
- For nonlinear boundaries, kernel SVMs are popular; kernels can be used with logistic regression and LDA as well, but computations are more expensive.

Summary

- A separating hyperplane classifies by $\text{sign} f(x)$; the maximal margin classifier maximizes $M = 1/\|\beta\|$ and depends only on support vectors.
- The support vector classifier allows margin violations with slacks under a budget C (chosen by CV).
- The solution depends on the data only through inner products, so kernels (polynomial, radial) give nonlinear SVMs without explicit feature expansion.
- OVA and OVO extend SVMs to $K > 2$ classes.
- SVM = hinge loss + ridge penalty, close to penalized logistic regression; use logistic regression when probabilities are needed.

7 Selected exercises

Motivation. ISLR Exercise 9.3 solves a maximal margin problem by hand, with a proof of optimality; the other two items derive the dual quoted in the classbox and prove that the radial kernel is a valid kernel.

Exercise 12.1 (A maximal margin classifier by hand; ISLR Ex. 9.3)

Seven points in $\mathbb{R}^2$: Red $(3,4), (2,2), (4,4), (1,4)$ (observations 1–4) and Blue $(2,1), (4,3), (4,1)$ (observations 5–7). Code Red as $y = +1$, Blue as $y = -1$.
(a) Find the optimal separating hyperplane and prove that it is optimal.
(b) State the classification rule, the margin and the support vectors.
(c) Show that a slight movement of observation 7 does not change the hyperplane.

Solution. We solve $\min \frac{1}{2}\|\beta\|^2$ subject to $y_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2}) \geq 1$.

(a)

Step 1 (a lower bound). The constraints of observations 2, 5 and 6 read

$$\beta_0 + 2\beta_1 + 2\beta_2 \geq 1, \qquad -\beta_0 - 2\beta_1 - \beta_2 \geq 1, \qquad -\beta_0 - 4\beta_1 - 3\beta_2 \geq 1.$$

Adding the first two gives $\beta_2 \geq 2$; adding the first and third gives $-2\beta_1 - \beta_2 \geq 2$, so $\beta_1 \leq -1 - \beta_2/2 \leq -2$. Hence every feasible β has $\|\beta\|^2 = \beta_1^2 + \beta_2^2 \geq 8$, with equality only for $\beta_1 = -2$, $\beta_2 = 2$.

Step 2 (the intercept). With $\beta_1 = -2$, $\beta_2 = 2$, the first constraint gives $\beta_0 \geq 1$ and the second $\beta_0 \leq 1$, so $\beta_0 = 1$.

Step 3 (feasibility). $f(x) = 1 - 2x_1 + 2x_2$ takes the values $3, 1, 1, 7$ at the Red points and $-1, -1, -5$ at the Blue points, so all constraints hold. By Steps 1–2 it is the unique optimum. The hyperplane is $1 - 2X_1 + 2X_2 = 0$, i.e. $0.5 - X_1 + X_2 = 0$.

(b) Classify to Red if $0.5 - X_1 + X_2 > 0$ and to Blue otherwise. The margin is $M = 1/\|\beta\| = 1/\sqrt{8} \approx 0.354$. The support vectors are the points with $y_i f(x_i) = 1$: observations 2, 3, 5 and 6.

(c) Observation 7 has $y_7 f(x_7) = 5$. If it moves by δ with $\|\delta\| < \sqrt{2}$, then $|f(x_7 + \delta) - f(x_7)| = |\beta^\top \delta| \leq \sqrt{8}\|\delta\| < 4$, so its constraint still holds and (β_0, β) remains feasible. The lower bound of Step 1 and Step 2 used only observations 2, 5 and 6, which did not move; so the same hyperplane is still the unique optimum. ■

Exercise 12.2 (The dual of the maximal margin problem)

For separable data, derive the dual of $\min \frac{1}{2}\|\beta\|^2$ subject to $y_i(\beta_0 + \beta^\top x_i) \geq 1$:

$$\max_{\alpha \geq 0} \sum_i \alpha_i - \frac{1}{2} \sum_{i,i'} \alpha_i \alpha_{i'} y_i y_{i'} x_i^\top x_{i'} \quad \text{s.t.} \quad \sum_i \alpha_i y_i = 0,$$

and show that $\hat{\beta} = \sum_i \hat{\alpha}_i y_i x_i$ with $\hat{\alpha}_i > 0$ only for points on the margin.

Solution [Supplement].

Step 1 (Lagrangian). For $\alpha \geq 0$ let $\mathcal{L}(\beta_0, \beta, \alpha) = \frac{1}{2}\|\beta\|^2 - \sum_i \alpha_i \{y_i(\beta_0 + \beta^\top x_i) - 1\}$ and $g(\alpha) = \inf_{\beta_0, \beta} \mathcal{L}$. For feasible (β_0, β) each brace is ≥ 0, so $g(\alpha) \leq \mathcal{L} \leq \frac{1}{2}\|\beta\|^2$ (weak duality).

Step 2 (the infimum). $\mathcal{L}$ is linear in β_0 with coefficient $-\sum_i \alpha_i y_i$, so $g(\alpha) = -\infty$ unless $\sum_i \alpha_i y_i = 0$. In that case $\mathcal{L}$ is a strictly convex quadratic in β with gradient $\beta - \sum_i \alpha_i y_i x_i$, minimized at $\beta = \sum_i \alpha_i y_i x_i$. Substituting, $\sum_i \alpha_i y_i \beta^\top x_i = \|\beta\|^2$, so

$$g(\alpha) = \frac{1}{2}\|\beta\|^2 - \|\beta\|^2 + \sum_i \alpha_i = \sum_i \alpha_i - \frac{1}{2}\Big\|\sum_i \alpha_i y_i x_i\Big\|^2,$$

which is the stated dual objective.

Step 3 (KKT). The primal is a convex quadratic program with linear constraints and is feasible, so by the standard duality theorem for such programs the optimal values agree and there is a dual optimum $\hat{\alpha}$ satisfying the KKT conditions with the primal optimum: stationarity $\hat{\beta} = \sum_i \hat{\alpha}_i y_i x_i$, $\sum_i \hat{\alpha}_i y_i = 0$, and complementary slackness $\hat{\alpha}_i \{y_i(\hat{\beta}_0 + \hat{\beta}^\top x_i) - 1\} = 0$.

Step 4. By complementary slackness, $\hat{\alpha}_i > 0$ forces $y_i f(x_i) = 1$: only points on the margin enter $\hat{\beta}$. For any such point s, $\hat{\beta}_0 = y_s - \hat{\beta}^\top x_s$ (as $y_s^2 = 1$), and $f(x) = \hat{\beta}_0 + \sum_i \hat{\alpha}_i y_i x_i^\top x$ uses the data only through inner products. ■

Exercise 12.3 (The radial kernel is positive semidefinite)

Show that for any $x_1, \dots, x_n \in \mathbb{R}^p$ and $\gamma > 0$ the matrix $K = \left[\exp(-\gamma \|x_i - x_j\|^2)\right]_{i,j}$ is positive semidefinite.

Solution [Supplement].

Step 1. The Gram matrix $G = [x_i^\top x_j]$ is PSD: $c^\top G c = \|\sum_i c_i x_i\|^2 \geq 0$.

Step 2 (Schur product). If A, B are PSD, so is the entrywise product $A \circ B$. Write $B = \sum_k \lambda_k v_k v_k^\top$ with $\lambda_k \geq 0$; then

$$c^\top (A \circ B) c = \sum_k \lambda_k \sum_{i,j} (c_i v_{ki}) A_{ij} (c_j v_{kj}) = \sum_k \lambda_k (c \circ v_k)^\top A (c \circ v_k) \geq 0.$$

Step 3. By induction, every entrywise power $G^{\circ m}$ is PSD ($G^{\circ 0} = \mathbf{1}\mathbf{1}^\top$). Hence $E_M = \sum_{m=0}^{M} \frac{(2\gamma)^m}{m!} G^{\circ m}$ is PSD, and $E_M \to E = \left[e^{2\gamma x_i^\top x_j}\right]$ entrywise; since $c^\top E_M c \geq 0$ for every M, also $c^\top E c \geq 0$.

Step 4. $\|x_i - x_j\|^2 = \|x_i\|^2 - 2x_i^\top x_j + \|x_j\|^2$ gives $K = DED$ with $D = \operatorname{diag}(e^{-\gamma \|x_i\|^2})$, so $c^\top K c = (Dc)^\top E (Dc) \geq 0$.

By the Mercer criterion quoted in the chapter, the radial kernel is an inner product in some (infinite-dimensional) feature space, as claimed. ■